



Assessment Task 2: Part B Group 3

Cluster Analysis Report: Customer Segmentation for Plant-Based Burger Preferences



24753 Customer Analytics - Autumn 2025

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Introduction

This research examines customer segmentation by conjoint-derived part-worth utility scores for preference in plant-based burgers. The clustering was done directly within Tableau with preference scores across various burger attributes, and two customer clusters were achieved.

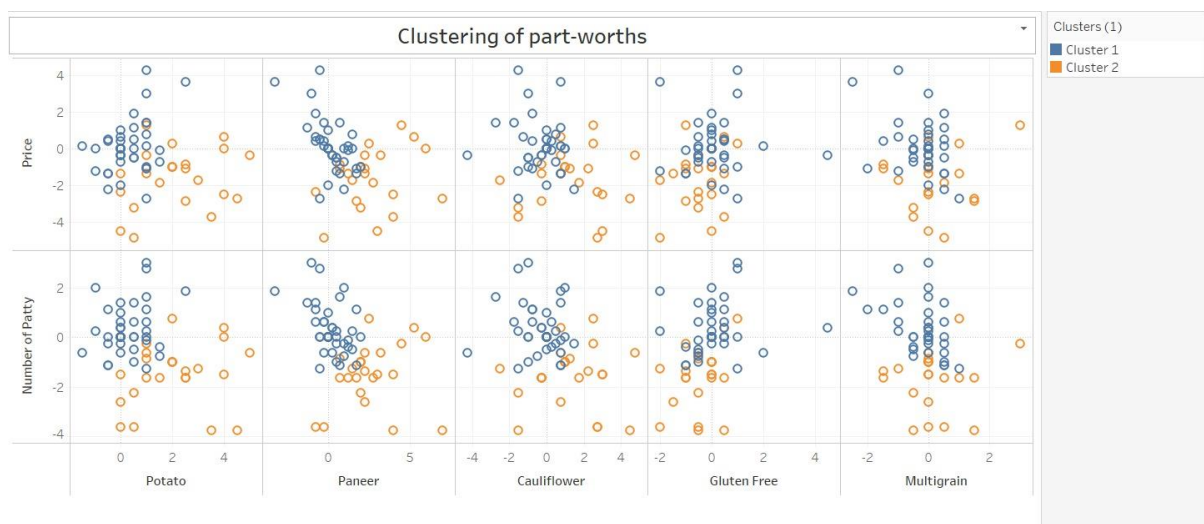
The data comes from a customer preference survey to obtain customer input on features for McDonald's future plant-based burger product lines. Features considered included patty type (Paneer, Potato, Cauliflower, Soy), bread (Multigrain), diet types (Gluten Free), price, and number of patties.

Clustering Methodology

The data set included part-worth scores at an individual level for the following attributes: Cauliflower, Gluten Free, Multigrain, Number of Patties, Paneer, Potato, and Price. These part-worth scores are respondent preferences estimated by the conjoint analysis.

These part-worth scores were fed into the clustering model in Tableau without aggregation. Tableau automatically called its internal K-Means clustering algorithm and decided on two clusters. There was no manual override to define the number of clusters.

Visualizations were created by plotting all the attributes against Price and Number of Patties for identifying cluster patterns. The respondents were color-coded by cluster to compare visually.



Cluster Profiles:

- **Cluster 1 (40 respondents):** More balanced in preferences. Shows moderate liking for common patties (Paneer and Potato), positive attitude toward Gluten Free options and more patties. Relatively neutral on price.



- **Cluster 2 (24 respondents):** Strong preference for Paneer, Potato, and Cauliflower. Prefers fewer patties and shows price sensitivity.

Inputs for Clustering

Variables: Sum of Cauliflower
Sum of Gluten Free
Sum of Multigrain
Sum of Number Patty
Sum of Paneer
Sum of Potato
Sum of Price

Level of Detail: Not Aggregated
Scaling: Normalized

Summary Diagnostics

Number of Clusters: 2
Number of Points: 64
Between-group Sum of Squares: 4.1559
Within-group Sum of Squares: 10.314
Total Sum of Squares: 14.47

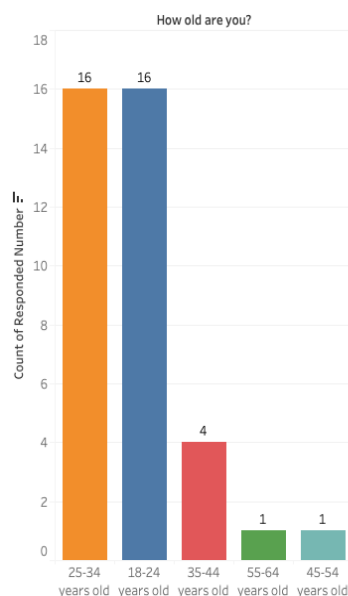
Clusters	Number of Items	Centers						
		Sum of Cauliflower	Sum of Gluten Free	Sum of Multigrain	Sum of Number Patty	Sum of Paneer	Sum of Potato	Sum of Price
Cluster 1	40	-0.30625	0.1	-0.1875	0.30938	0.19375	0.2875	0.10312
Cluster 2	24	1.25	-0.41667	0.16667	-1.5	2.5833	2.0833	-1.5833
Not Clustered	0							

The values shown in the cluster centers are the mean part-worth scores for each attribute in each cluster. The higher values represent greater preference for the feature, and the lower (especially negative) values represent lower interest or aversion.

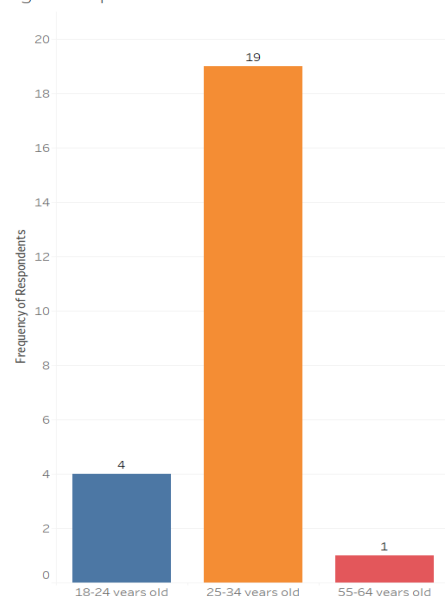
Cluster Demographic Comparison:

1. Age Group

Age Group



Age Group

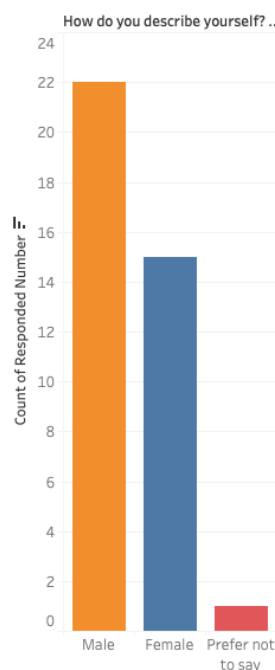




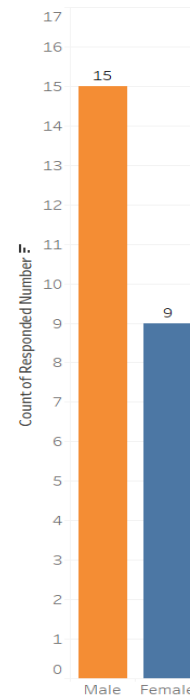
- **Cluster 1** is composed almost equally of respondents aged **18–24 and 25–34**, reflecting a wider appeal among young adult consumers.
- **Cluster 2** is **more concentrated in the 25–34** age group, with fewer younger consumers.
- Other age groups such as 35–44, 45–54, and 55–64 are seen marginally in both the clusters.

2. Gender Distribution

Gender



Gender Composition



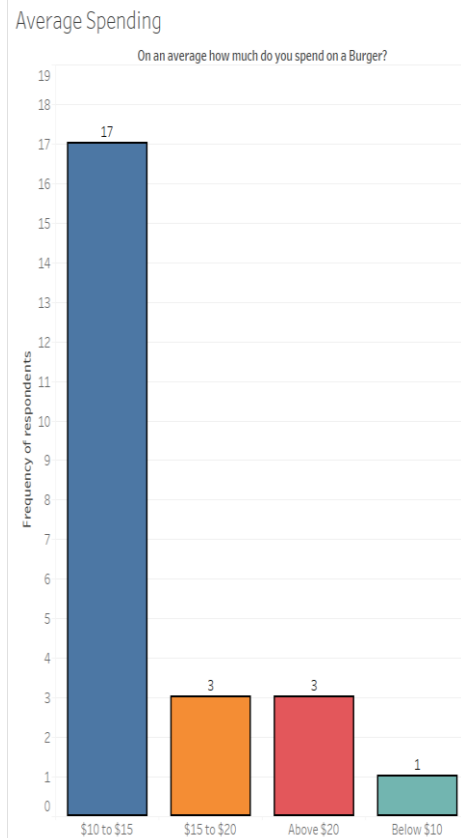
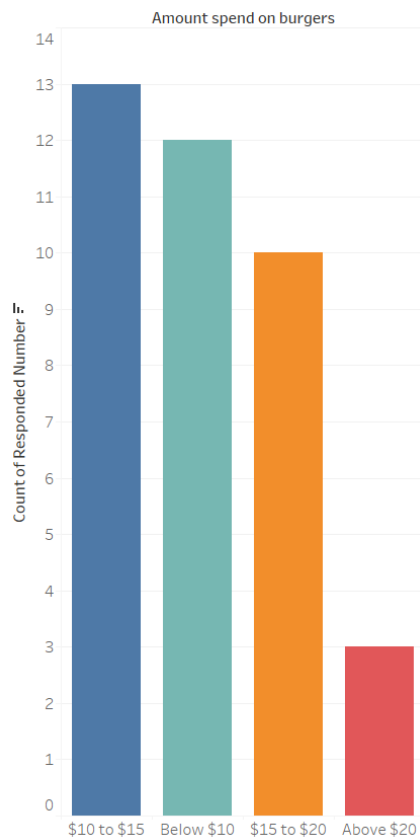
- **Cluster 1** consists of 15 males, 9 females, and 1 preferring not to say.
- **Cluster 2** contains a larger male population with 22 males and 15 females, which reflects skew towards male preference for plant over animal choices.

Insight: Both clusters are male dominant, but to a much greater degree in Cluster 2.



Consumer Behaviour Analysis

3. Spending Patterns

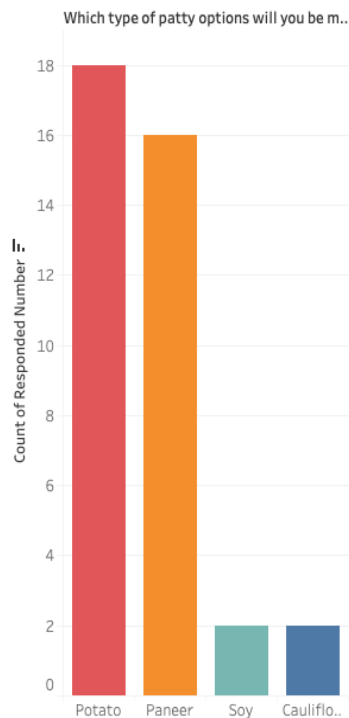


- **Cluster 1** spends predominantly in the **\$10–\$15** range, showing consistency and medium price tolerance.
- **Cluster 2** is more varied, with presence in **below \$10 and above \$20** segments, indicating the existence of both budget and premium consumers.

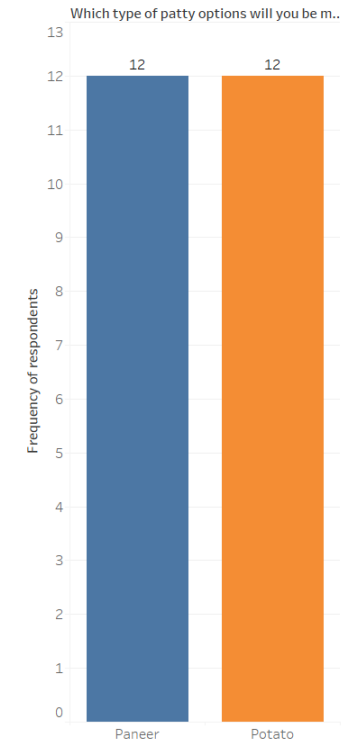


4. Patty Preference

Patty Preference



Patty Preference



- **Cluster 1** respondents shared an equal response between Paneer and Potato patties, with 13 each.
- **Cluster 2** showed a strong preference towards Potato (18), then Paneer (16), with weak interest in Soy and Cauliflower.

Insight: Both groups prefer classic veggie options (Paneer and Potato), even though Cluster 2 shows some stronger inclination toward accepting different kinds of patty.



Recommendations to McDonald's:

Based on these cluster profiles, McDonald's can follow a dual-target approach:

Target Segment 1: "Value-Focused Millennials" (Cluster 1)

- Who they are: Predominantly 18–24 and 25–34-year-olds, price-sensitive, prefer traditional and familiar vegetarian flavours (Paneer, Potato).
- Product Fit: Offer base burger combos with Paneer and Potato patties
- Offer Gluten Free and Multigrain bread as upselling opportunities
- Set price points between \$10–\$15 combos
- Promote "value meals" with homogenous Flavors
- Marketing Channel: Ads targeting students, app coupons, campus campaigns, budget meal promotions.

Target Segment 2: "Experimental Urban Adults" (Cluster 2)

- Who they are: Primarily 25–34 years old, male-skewed, willing to experiment with premium patties (Cauliflower, Soy), more diversified in price aspirations.
- Product Fit: Roll out limited-time burgers featuring unconventional patties like Cauliflower.
- Introduce customization options for patty size and variety (e.g., single patty, gourmet toppings)
- Create premium SKUs over \$15 but also offer a base model under \$10
- Marketing Channel: Digital channels (Instagram, YouTube), influencer marketing, focus on sustainability and ingredients narrative.